

THAMES TOWER, United Kingdom

WMO: 036950

Lat: 51.667N

Lon: 1.100E

Elev: 26

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 82-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.1	33.2	23.7	17.9	33.0	27.4	21.3	36.2	45.7	38.8	43.2	39.0	23.2	70	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	71.0	66.7	69.6	65.8	68.2	64.8	68.0	69.9	66.7	68.7	65.5	67.7	12.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	100.3	69.4	65.9	95.6	68.2	64.5	91.0	67.3	32.3	70.2	31.3	68.8	30.4	67.8	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
35.3	31.7	28.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	42.4	41.8	44.6	47.2	52.8	58.0	63.0	64.0	59.7	54.8	48.1	45.3
	DBStd	8.70	4.85	5.04	3.89	3.85	4.00	3.55	3.09	3.43	3.29	4.19	4.95	4.85
	HDD50	1018	237	230	170	99	19	0	0	0	0	11	94	158
	HDD65	4837	701	650	632	534	378	211	77	60	161	315	508	610
	CDD50	1706	1	1	3	16	106	240	404	435	290	161	36	13
	CDD65	49	0	0	0	0	0	1	16	30	2	0	0	0
	CDH74	4	0	0	0	0	0	0	2	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.1	17.6	16.2	14.3	14.6	14.4	13.9	12.9	14.2	15.2	16.5	14.9	16.7
Precipitation	PrecAvg	22.3	2.0	1.6	1.3	1.5	1.7	2.0	1.8	2.1	1.9	2.4	2.3	2.1
	PrecMax	29.5	4.2	3.7	3.5	3.9	3.8	4.8	3.7	4.5	3.9	5.8	4.6	4.1
	PrecMin	15.3	0.4	0.2	0.4	0.0	0.3	0.4	0.4	0.2	0.3	0.0	0.7	0.5
	PrecStd	3.7	0.9	0.9	0.8	1.0	1.0	1.3	0.8	1.3	1.0	1.3	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	52.8	54.8	58.0	63.0	68.8	73.0	73.8	69.3	65.0	59.9	56.7
		MCWB	51.4	50.6	51.9	54.7	59.4	64.6	66.6	68.8	63.5	61.5	58.5	55.6
	2%	DB	51.3	51.1	52.9	55.6	60.9	66.2	70.6	71.5	67.0	62.7	58.0	55.3
		MCWB	49.5	49.0	50.3	53.3	58.0	62.6	66.0	67.3	62.5	59.9	56.7	54.6
	5%	DB	49.9	49.8	51.4	53.9	59.5	64.5	69.0	70.2	65.8	61.2	56.3	54.0
		MCWB	48.3	47.7	49.2	51.6	56.9	61.4	64.9	66.5	61.8	58.6	54.9	53.1
	10%	DB	48.5	48.3	50.0	52.5	58.2	62.9	67.6	69.0	64.5	60.1	54.8	52.3
		MCWB	47.0	46.5	48.1	50.5	55.8	60.3	64.0	65.7	61.0	57.6	53.1	51.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.5	52.6	55.8	60.3	65.3	69.3	70.3	65.0	62.4	59.2	56.4
		MCDB	53.2	52.3	54.2	57.1	62.3	67.5	70.7	72.4	66.8	63.7	59.5	56.6
	2%	WB	50.0	49.5	50.9	53.7	58.5	63.3	67.4	68.8	63.1	60.6	57.3	55.0
		MCDB	51.1	50.7	52.3	54.9	60.2	65.1	69.1	70.7	65.0	61.8	57.9	55.4
	5%	WB	48.6	48.3	49.7	52.1	57.5	61.9	65.8	67.4	62.2	59.4	55.4	53.5
		MCDB	49.7	49.6	51.1	53.3	59.1	63.7	67.7	69.4	64.0	61.0	56.2	54.0
	10%	WB	47.5	46.9	48.5	50.9	56.3	60.7	64.7	66.5	61.2	58.2	53.3	51.5
		MCDB	48.5	48.4	49.8	52.3	58.1	62.5	66.9	68.6	63.0	59.9	54.6	52.3
Mean Daily Temperature Range	5% DB	MDBR	5.2	4.9	5.0	5.4	5.1	5.9	6.3	6.2	5.7	5.4	4.9	5.1
		MCDBR	7.1	6.1	6.7	6.9	6.5	7.9	7.9	7.5	6.5	5.9	4.4	5.8
		MCWBR	7.1	5.8	6.1	5.6	5.1	6.2	6.0	5.9	5.1	5.1	4.2	5.5
	5% WB	MCDBR	6.6	5.9	6.6	6.6	6.1	7.2	6.9	6.4	5.8	5.2	4.1	5.1
		MCWBR	6.8	5.8	6.3	5.6	5.1	6.2	5.9	5.4	5.3	5.0	4.1	5.5
Clear-Sky Solar Irradiance	taub	0.331	0.348	0.384	0.405	0.412	0.414	0.411	0.403	0.382	0.365	0.338	0.323	
	taud	2.381	2.386	2.280	2.254	2.282	2.301	2.318	2.364	2.412	2.433	2.421	2.373	
	Ebn at Noon	213	244	254	263	267	267	266	262	254	234	209	195	
	Edh at Noon	20	25	34	39	40	39	38	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg	245	458	813	1302	1552	1711	1626	1357	1018	584	297	190	
	RadStd	23	69	100	125	112	133	148	103	82	55	34	25	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page